LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART II

SESSION 9

Fiscal Policy, Debt & Sovereign Risk



From markets, to macro, to decisions that matter — in Nigeria and the world.

Reading: Net Fiscal Operations · The Fiscal Responsibility Law (CBN Brief) · Facilitator: Elvis Ozegbe

RECAP

Session 8 in Three Ideas

Policy rate

The CBN sets the price of overnight money; banks, bonds and the naira react within days.

Transmission

Rates reach your firm through credit costs, asset prices, the exchange rate and expectations — with long lags.

Fiscal dominance

When the central bank must finance the government, monetary policy loses its grip. Today we look at the government side.

TODAY'S PLAN

Session 9 at a Glance — 75 Minutes

How governments raise, spend and borrow — and when debt becomes a business risk.



Recap & Hook

Is the debt the problem — or the revenue?

The Fiscal Picture

Revenue, spending, deficits and the interest bill

Debt Dynamics

r, g, the primary balance — and three futures

Lab & Discussion

Project Nigeria's debt; what it means for your firm

BY THE END OF TODAY

Session 9 Learning Objectives

You will be able to:

01

Read a Budget Like an Analyst

Separate revenue, primary spending and interest, and judge the burden against revenue as well as GDP.

02

Use the Debt-Dynamics Equation

Explain why debt/GDP rises or falls using r, g and the primary balance — and spot the 'other' items.

03

Turn Fiscal Risk Into Business Risk

Link subsidy reform, tax reform, borrowing and sovereign risk to rates, the naira and demand in your market.

THE HOOK

Low Debt, Heavy Burden

“Nigeria's debt is only about 40% of GDP — Japan's is over 200%. What is the fuss about?”

A treasurer at a Lagos bank asks three better questions:

₵ What can the state collect?

Debt is repaid from revenue, not GDP. FGN revenue is only about **5% of GDP** in our data.

% What does the debt cost?

Interest absorbed **two-thirds of revenue** in 2025 and over **90%** in 2021.

⇄ In which currency?

Dollar debt grows overnight in naira when the exchange rate moves — as it did in 2023–24.

NIGERIAN BUSINESS CASE

Is the Debt the Problem — or the Revenue?

The situation

Federal debt rose sharply after 2022 as the CBN overdraft was converted into bonds and the naira fell. Interest payments now take most of federal revenue, and banks hold large amounts of FGN bonds.

The decision question

Is Nigeria's fiscal problem **too much debt**, or **too little revenue** — and what does each answer imply for interest rates, the naira and your company's plans?

The data we have

fgn_fiscal_annual.csv — 2010–2025: nominal GDP, FGN revenue (and oil share), spending, interest, total public debt. *Stylised teaching data.*

What a good answer looks like

A short diagnosis using **three ratios** (debt/GDP, debt/revenue, interest/revenue), a **debt-dynamics** decomposition and a ten-year projection under three scenarios.

PART

01

The Fiscal Picture

Where the money comes from, and where it goes.

Four Numbers in Every Budget

₵ Revenue

Taxes (company income tax, VAT, customs), oil and gas receipts and other income. Nigeria's oil share of federal revenue has fallen but is still large.

⌂ Primary spending

Everything except interest: salaries, capital projects, transfers, subsidies. This is what the government chooses each year.

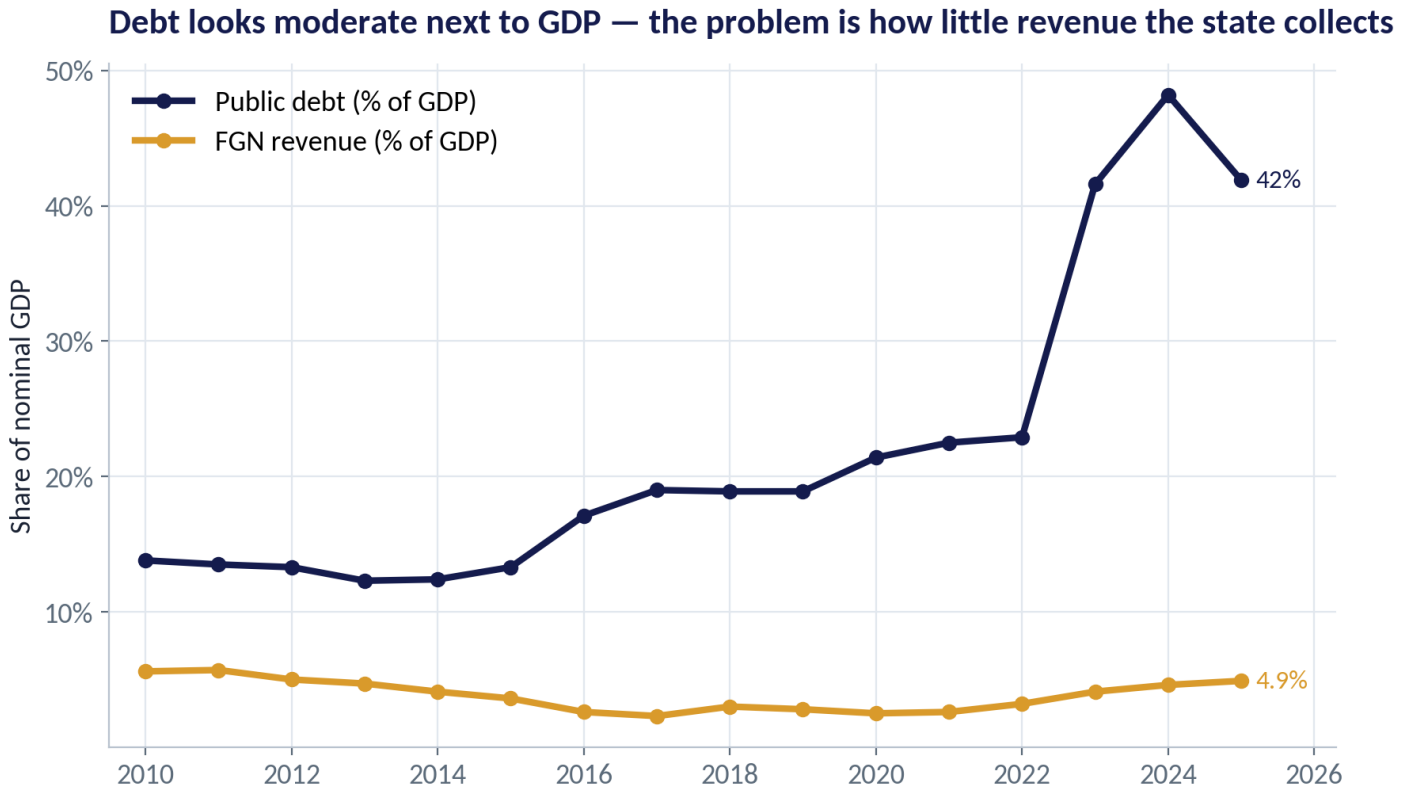
% Interest

The cost of past borrowing. It is not a choice this year — it is owed.

⚖ Primary balance

Revenue minus primary spending. A primary **surplus** means the state can pay some interest from its own income.

Moderate Debt, Very Low Revenue



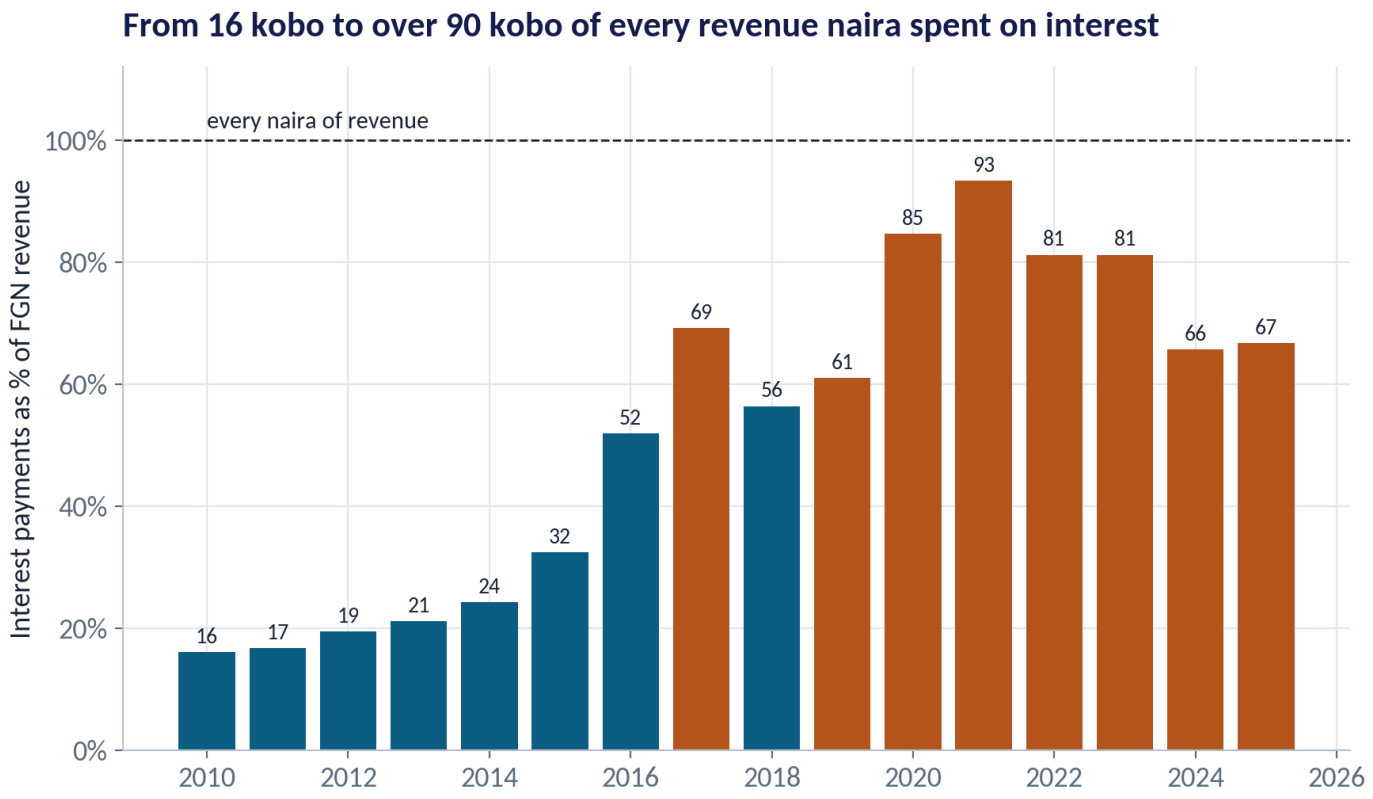
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S09-F01 · replicate: s09_figures.R / .py

WHAT TO SEE

- Public debt rose from **14% of GDP in 2010 to 42% in 2025** in our stylised data.
- FGN revenue stayed around **3–5% of GDP** — among the lowest ratios in the world for a large economy.
- A debt ratio that is comfortable for a country collecting 20–30% of GDP is heavy for one collecting 5%.
- So judge the debt against **what the state can collect**, not only against the size of the economy.

How Much Revenue Goes on Interest?



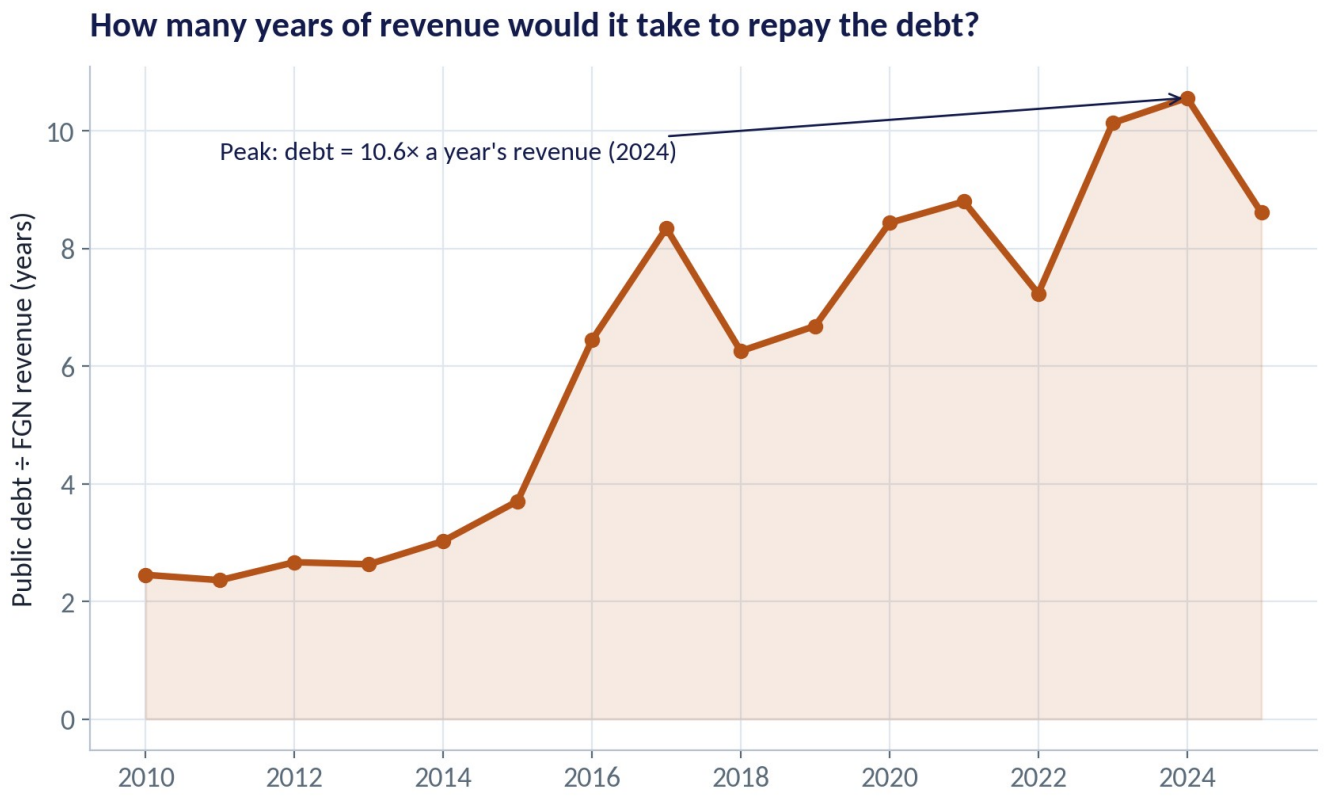
Source: EEGB stylised teaching data (data/). Not official statistics.

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WHAT TO SEE

- In 2010 interest took about **16 kobo** of every naira of federal revenue.
- By 2021 it took **93 kobo**, leaving almost nothing for salaries, roads or schools without more borrowing.
- Higher oil receipts and reforms lowered the ratio to about **67%** in 2025 — still very high.
- Published reports showed a similar pattern; our numbers are stylised.

Years of Revenue Needed to Repay the Debt



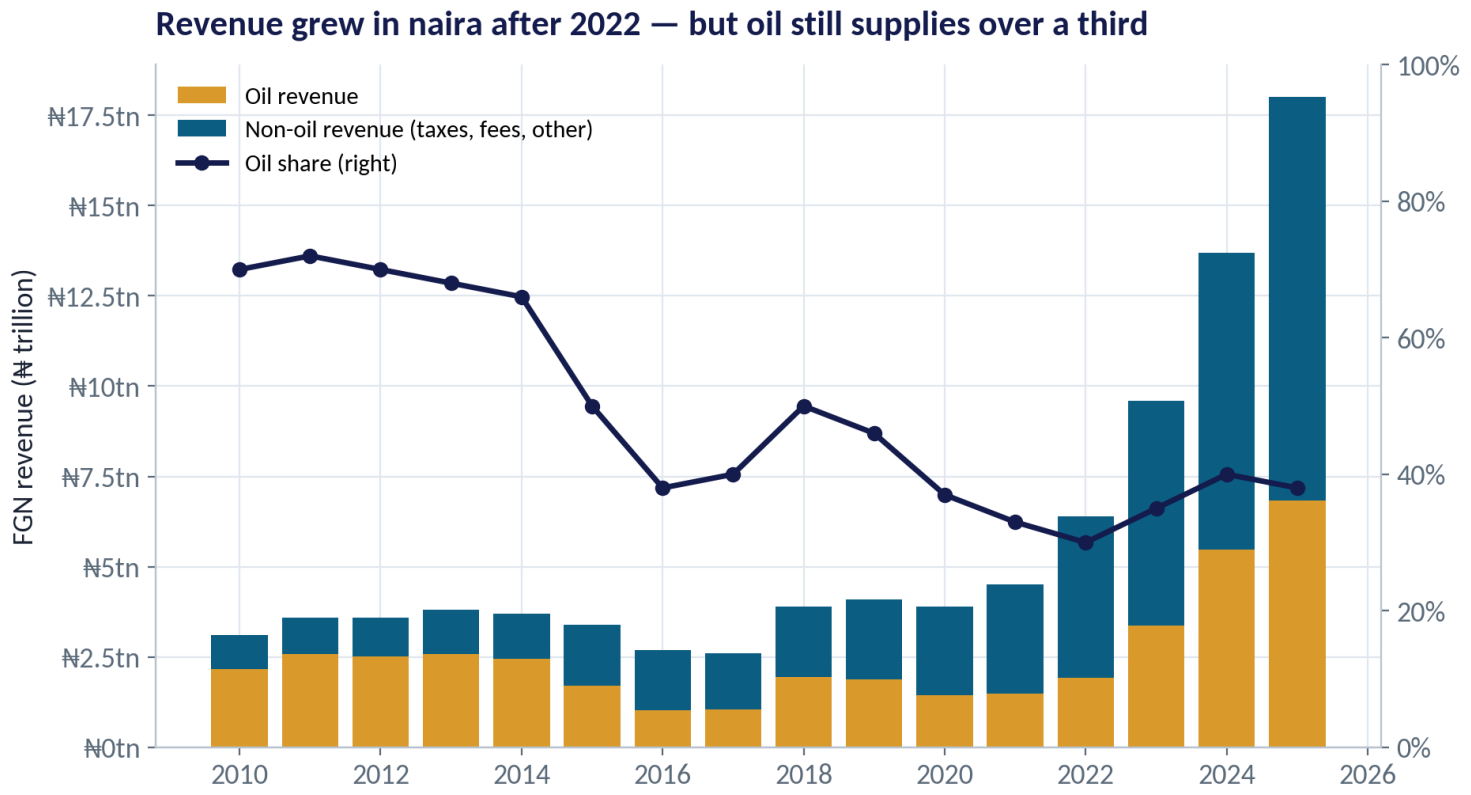
Source: EEGB stylised teaching data (data/). Not official statistics.

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WHAT TO SEE

- Debt ÷ revenue: how many years of **all** federal revenue would repay the stock of debt.
- It rose from about **2.5 years in 2010** to a peak of **10.6 years in 2024**.
- By 2025 it eased to **8.6 years** as naira revenue rose faster than debt.
- This is the number that explains why investors demand high yields on FGN debt.

Revenue: Bigger in Naira, Still Oil-Heavy



Source: EEGB stylised teaching data (data/). Not official statistics.

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WHAT TO SEE

- Oil supplied about **70%** of federal revenue in the early 2010s; under **40%** in recent years.
- Naira revenue jumped after 2022 — partly because a weaker naira raises the naira value of oil sales.
- Oil prices and output still swing the budget, so **fiscal risk is also commodity risk**.
- The 2025 tax reform laws aim to raise non-oil revenue; firms should expect tighter tax administration.

Quick Check: What Would You Say?

?

Country A has debt of 40% of GDP and revenue of 5% of GDP. Country B has debt of 80% of GDP and revenue of 35% of GDP. Which is more stretched, and why?

A. Country B — its debt ratio is twice as high

B. Country A — its debt equals 8 years of revenue versus about 2.3 years in B

C. They are equally stretched

D. Neither — debt ratios below 100% are always safe

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

02

Debt Dynamics

One equation explains why debt ratios rise or fall.

THE KEY EQUATION

What Moves the Debt Ratio?

IN PLAIN ENGLISH

The debt ratio rises when the **interest rate (r)** on debt is higher than the economy's **nominal growth (g)**, and when the government runs a **primary deficit**. Anything else — a devaluation, a bailout, an old overdraft turned into bonds — shows up as **'other'**.

EXAMPLE

In 2025 in our data: r = **8.3%**, g = **23.3%**. With g far above r, the interest–growth term cut the debt ratio by about **6 points** on its own.

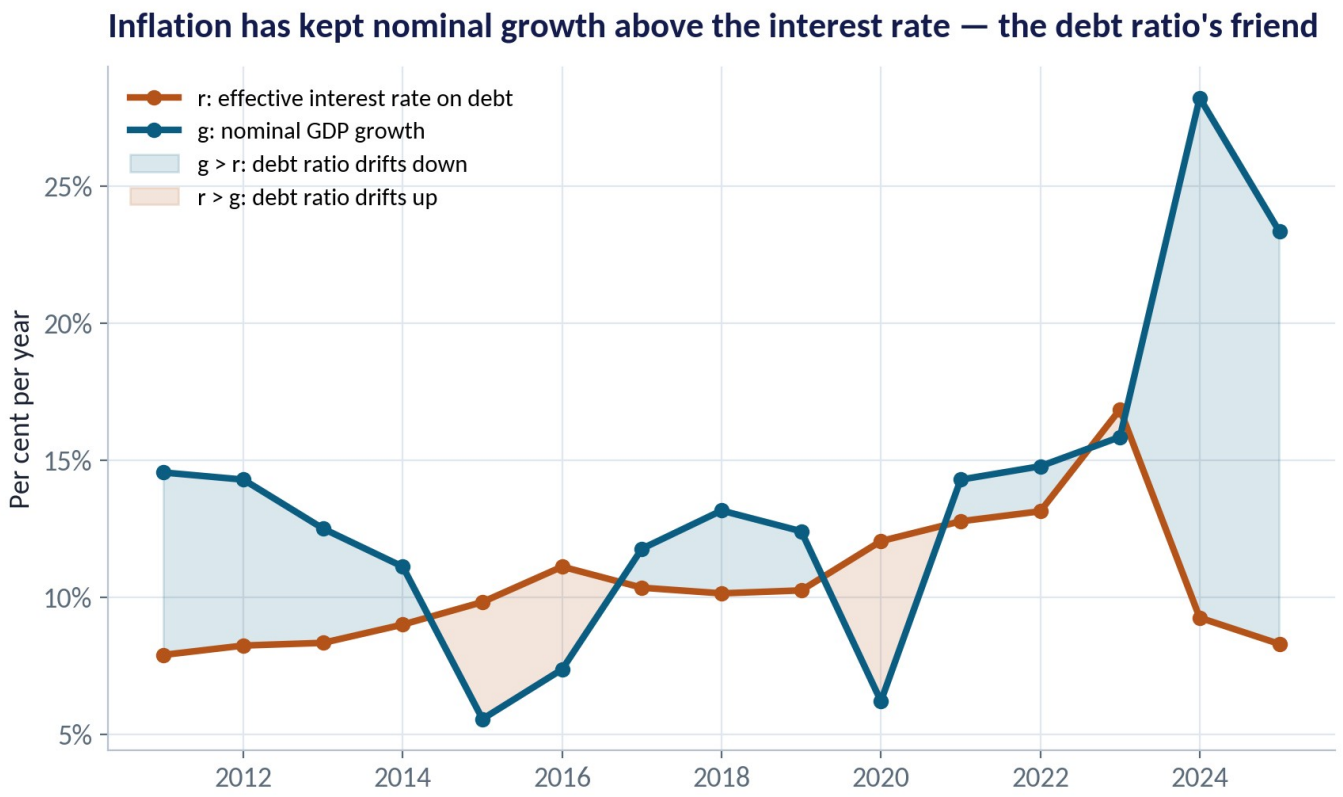
THE FORMULA

$$\Delta d = (r - g) \div (1 + g) \times d \; - \; pb \; + \; other$$

WHY A MANAGER CARES

High inflation raises g and quietly erodes the debt ratio — but it also raises future interest rates. The relief lasts only while r stays below g.

Inflation Has Been the Debt Ratio's Friend



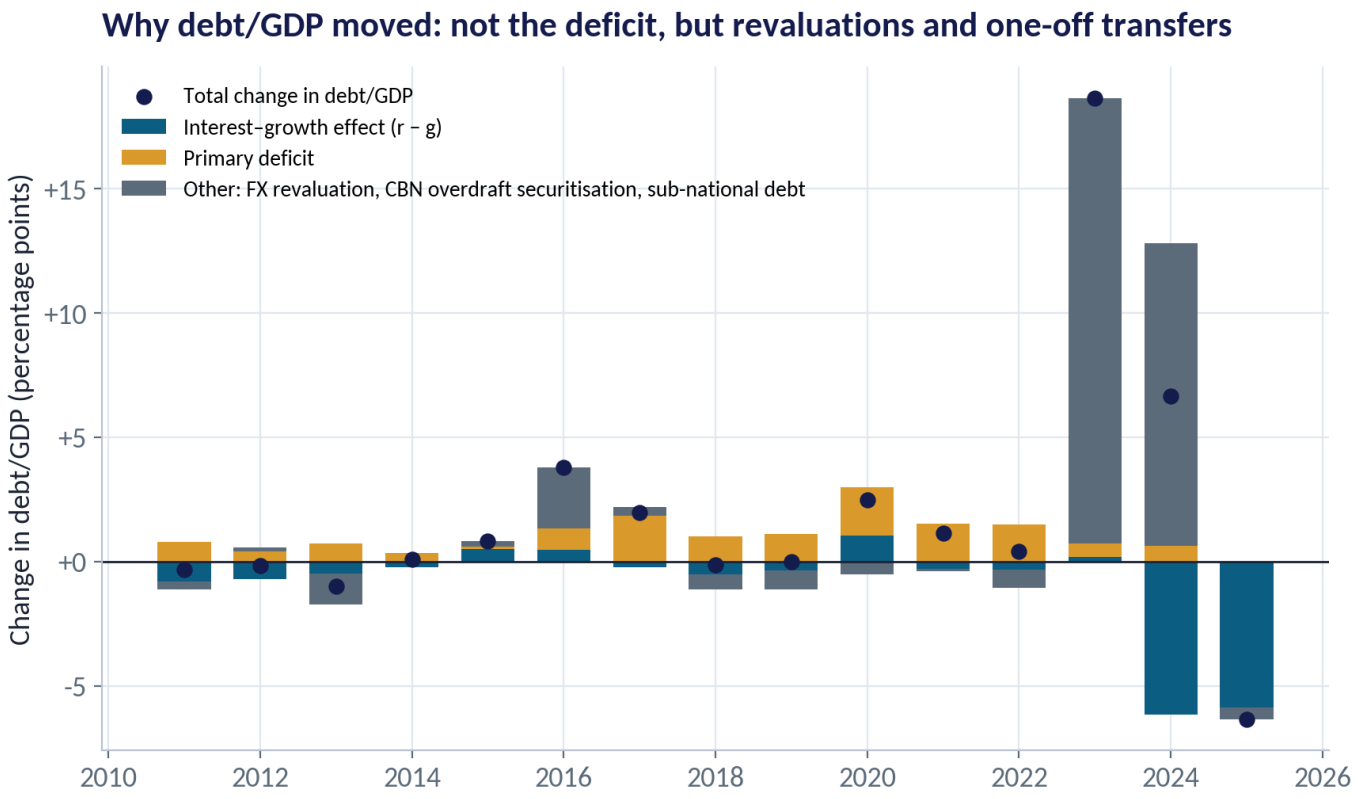
Source: EEGB stylised teaching data (data/). Not official statistics.

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WHAT TO SEE

- **g** (nominal growth) includes inflation; **r** is the average interest rate the state pays.
- In most years in our data g exceeded r, so the interest–growth term pulled the ratio **down**.
- When inflation falls, g falls quickly; r falls only as old debt is refinanced.
- Lesson: disinflation without fiscal reform can make the debt arithmetic **worse**.

Why Did Debt/GDP Jump in 2023–24?



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S09-F05 · replicate: s09_figures.R / .py

WHAT TO SEE

- 2023: debt/GDP rose **18.7 points** — almost all of it (**17.9**) in 'other'.
- That reflects the CBN overdraft ('Ways and Means') converted into bonds and the **naira revaluation of dollar debt**.
- 2024: 'other' added **12.2 points** while fast nominal growth subtracted **6.1**.
- Deficits matter, but the big jumps came from **off-budget items and the exchange rate**.

WORKED EXAMPLE

What Primary Balance Stabilises the Debt?

Set Δd = 0 and solve: pb = (r − g) ÷ (1 + g) × d*

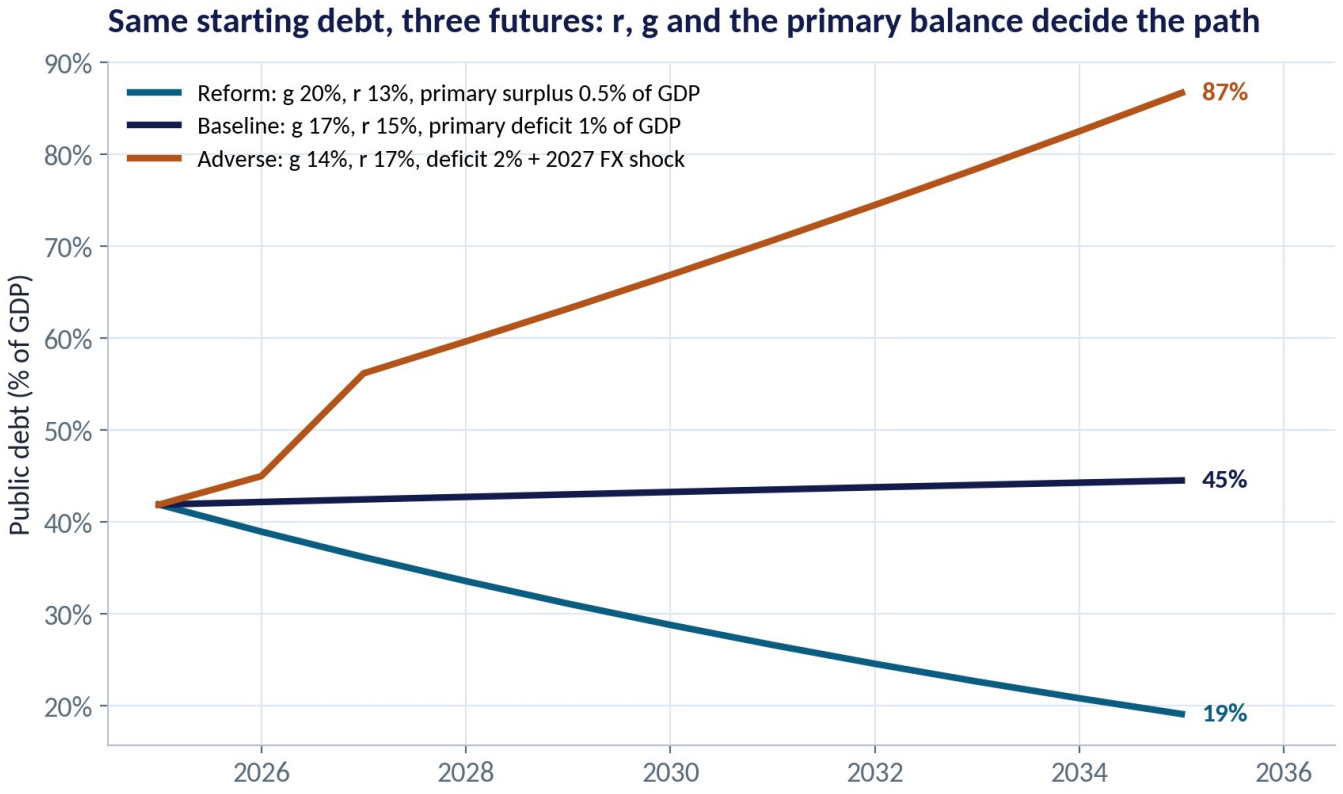
r	g	d	pb* (% of GDP)
15%	17%	41.9%	−0.7
17%	14%	41.9%	+1.1
20%	12%	41.9%	+3.0

STEP BY STEP

1. Start from 2025's debt ratio, **41.9% of GDP**.
2. If g beats r by 2 points, a small primary **deficit** of 0.7% of GDP still keeps the ratio flat.
3. If r beats g by 3 points, the state needs a primary **surplus** of 1.1% of GDP just to stand still.
4. With r 8 points above g, the required surplus is **3% of GDP** — over half of current revenue.

The same debt can be safe or explosive. **r − g** decides how much fiscal effort is needed.

A Ten-Year Projection



Source: EEGB stylised teaching data (data/). Not official statistics.

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WHAT TO SEE

- **Reform** (g 20%, r 13%, surplus 0.5%): debt falls to about **19% of GDP** by 2035.
- **Baseline** (g 17%, r 15%, deficit 1%): roughly flat at **45%**.
- **Adverse** (g 14%, r 17%, deficit 2%, an FX shock in 2027): debt more than doubles to **87%**.
- The difference comes from assumptions a CFO can watch: inflation, yields, the deficit and the naira.

Quick Check: What Would You Say?

?

Inflation falls sharply, so nominal GDP growth drops from 23% to 13%, while the average interest rate on government debt stays at 15%. Holding the primary balance constant, what happens to the debt ratio?

A. It falls, because lower inflation is good news

B. It starts rising, because r is now above g

C. Nothing — inflation does not affect debt

D. It falls, because interest rates will fall immediately

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

03

Fiscal Policy and Your Business

Multipliers, rules, reforms and sovereign risk.

Fiscal Levers and What They Mean for Firms

Lever	Nigerian example	Effect on firms
Subsidy reform	Petrol subsidy removal (2023)	Higher transport and energy costs now; more fiscal space later if savings are spent well
Tax reform	2025 tax reform laws; VAT and company tax changes	Compliance costs, possible relief for small firms, tighter enforcement for large ones
Borrowing	Large domestic bond issuance	High yields crowd out private credit; banks prefer FGN bonds to loans
Central-bank financing	Ways and Means advances (before 2023)	Money growth, inflation and a weaker naira
Fiscal rules	Fiscal Responsibility Act 2007 (deficit cap of 3% of GDP)	Rules anchor expectations only if they are respected

Does Government Spending Grow the Economy?

IN PLAIN ENGLISH

A **fiscal multiplier** tells you how much GDP changes when government spending changes by ₦1. It is larger when the economy has idle capacity and smaller when spending leaks into imports or pushes up interest rates.

EXAMPLE

If ₦1 trillion of road spending raises GDP by ₦0.6 trillion, the multiplier is **0.6**. Open, import-dependent economies with high interest rates tend to have multipliers **below 1**.

THE FORMULA

$$\text{multiplier} = \Delta\text{GDP} \div \Delta\text{G}$$

WHY A MANAGER CARES

When government cuts spending, your demand falls by more in sectors that sell to the state (construction, contractors) and less in export or consumer-staples sectors. Know where your revenue sits.

How Fiscal Stress Reaches Your Firm

Through the banks

The 'sovereign–bank doom loop'

- Banks hold large amounts of FGN bonds
- If bond prices fall, bank capital falls (Session 10)
- Banks cut lending to firms
- Credit becomes scarcer and dearer

Through markets

Yields, ratings and the naira

- Higher sovereign yields raise every borrower's cost
- Rating downgrades raise Eurobond spreads
- Foreign investors pull money; the naira weakens
- Tax rises or spending cuts hit demand

A company is rarely rated above its sovereign. Your cost of capital starts from the government's.

How Managers Misread Fiscal News

THE TRAP	THE BETTER HABIT
✗ “Debt is only 40% of GDP, so it is safe.”	✓ Compare debt and interest with **revenue**, and check the **currency** of the debt.
✗ Treating a big naira revenue increase as fiscal strength.	✓ Ask how much is **devaluation and inflation** — real revenue may barely move.
✗ Assuming lower inflation automatically eases the debt burden.	✓ Check **r − g**: disinflation can raise the real burden of debt.
✗ Ignoring off-budget items.	✓ Watch for **'other'**: overdraft conversions, guarantees, state debt and FX revaluation.
✗ Planning on the budget's capital spending as announced.	✓ Track **actual releases**; in revenue-poor years capital budgets are cut first.

HANDS-ON LAB

Lab: Is the Debt the Problem — or the Revenue?

YOU WILL NEED

Data
`data/fgn_fiscal_annual.csv`

R script
`sessions/s09/R/s09_lab.R`

Python script
`sessions/s09/python/s09_lab.py`

⏱ **15 minutes**

1. Compute **debt/GDP, debt/revenue and interest/revenue** for every year.
2. Compute **r** (interest ÷ last year's debt) and **g** (nominal GDP growth). In which years was g > r?
3. Decompose each year's change in debt/GDP into **r − g, primary balance and 'other'**.
4. Project debt/GDP to 2035 under a **baseline and a reform** scenario.
5. Find the **debt-stabilising primary balance** for three combinations of r and g.

Deliverable: a 150-word note to your board: is the problem debt or revenue, and what should we watch?

WORDS TO OWN

Today's Vocabulary in Plain English

Primary balance	Revenue minus all spending except interest.
Fiscal deficit	Total spending (including interest) minus revenue.
Debt-to-GDP	Public debt as a share of the economy's annual output.
Interest-to-revenue	Share of government revenue spent on interest.
r − g	Interest rate on debt minus nominal growth; drives the debt ratio.
Stock-flow adjustment	Changes in debt not explained by the deficit (FX, bailouts, conversions).
Fiscal multiplier	Change in GDP per naira of extra government spending.
Fiscal dominance	Monetary policy constrained by the need to finance the government.
Ways and Means	Central-bank overdraft to the government.
Sovereign spread	Extra yield investors demand on a country's debt over a safe benchmark.

KEY TAKEAWAYS

Session 9 in 5 Ideas

01 Judge debt against **revenue and interest**, not only GDP.

02 Debt dynamics: **r − g** and the **primary balance** drive the ratio; 'other' items cause the jumps.

03 Inflation and devaluation cut both ways: they raise naira revenue and **naira-valued dollar debt**.

04 Fiscal stress reaches firms through **banks, yields, the naira and demand**.

05 Watch the few numbers that decide the path: **yields, inflation, the deficit and the exchange rate**.



NEXT › Session 10: Financial Stability

Before next session: Read the Silicon Valley Bank case (F-0997-E). Ask your treasury team how much of your bank's or company's investments are in long-dated FGN bonds, and at what yield they were bought.